❖ Site plan

The full-scale drawing indicating the architectural features of a location is known as a site plan**.**

A site plan is a detailed architectural drawing that exhibits the form, location, and orientation of a building on a site.

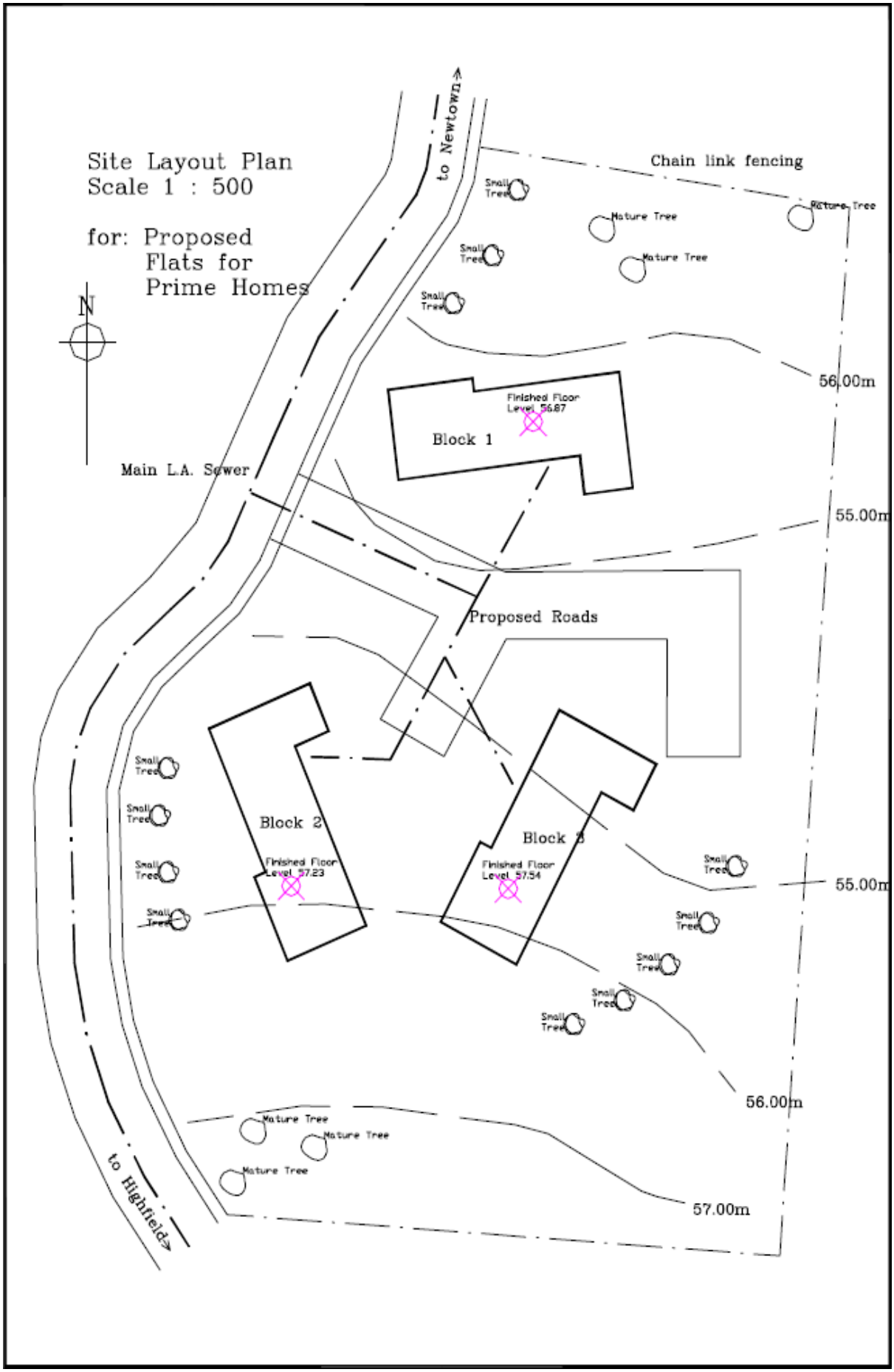
It generally includes the dimensions, contours, landscapes, and significant features of a piece of land and is also commonly referred to as the plot plan.

Such a plan is a graphic representation of the various aspects of a development project which includes the arrangement of buildings, driveways, parking facilities, water lines, lightings, and parks etc. It is further used by the developers to make advancements to a property.

The extent of the information shown on a site plan generally depends on the size and the nature of the project.

Certain information is likely to appear on most site plans:

- Title block, mentioning the name of the project, the type of drawing, the author, the revision number etc
- Directional orientation
- Major dimensions
- Key materials
- Site boundaries including the adjoining or adjacent boundaries
- Location of the building
- Parking areas
- The layout of external services like drainage, water, gas, and electricity
- Roads, footpaths, ramps and so on


Site Layout Plan
Scale 1 : 500
for: Proposed
Flats for
Prime Homes
N
to Newtown
Chain link fencing
Small Tree
Mature Tree
Mature Tree
Small Tree
Mature Tree
Small Tree
56.00m
Finished Floor Level 56.87
Block 1
Main L.A. Sewer
55.00m
Proposed Roads
Small Tree
Small Tree
Small Tree
Small Tree
Block 2
Finished Floor Level 57.23
Block 3
Finished Floor Level 57.54
Small Tree
55.00m
Small Tree
Small Tree
Small Tree
Small Tree
56.00m
Mature Tree
Mature Tree
Mature Tree
57.00m
to Highfield

❖ Selection of site for Buildings

Selection of site for building construction is important because it help to construct. Building Sizes use based on all factors such as structural, Soil, Environmental. Factory buildings always planned based on their function which made construction very simple.

For planning and designing of a building in best and cheap way proper selection of site is important. It is rarely possible to get a site which will meet all the criteria listed below. But site should be so selected that it satisfies the following criteria to a greater extent.

1. Site should serve the intended purpose and scope of the building to get constructed.

2. Size, shape, elevation, and ground conditions.

(a) Site should be uniform in geometrical shape since with irregular shape land is wasted.

(b) Site should be spacious enough to build for project and future requirements.

(c) The site should be flat and should not be in depression. Problem of drainage and sanitation becomes difficult due to which plinth height need to provide at design height.

(d) To reduce cost of foundation, hard and firm strata should be available at a shallow depth. Reclaimed, marshy, waterlogged, and black cotton soil, strata should be avoided.

3. Facilities: To the extent possible following facilities should be in proximity:

(a) Transport, electricity, water, school, shopping centres, post and telecommunications, cooking gas, hospitals.

(b) Site should be safe and in well secured locality.

4. Ownership: Site should be free from legal litigations.

5. Site near river, nallah should be above the highest flood level of nearby river area.

6. It should free from pollution due to smoke, dust, noise, insanitary conditions.

❖ Layout of Building

The real meaning and purpose of setting out (layout) is to transfer the plan, length, and width of its foundation on the ground so that the foundation can be excavated for construction of purposed building as per drawing.

The following preliminary works should be executed before actual planning of layout for the house.

- Clear the site from all grass, bushes, trees, etc.
- Record spot levels of the ground.
- Construct a permanent benchmark in construction area.

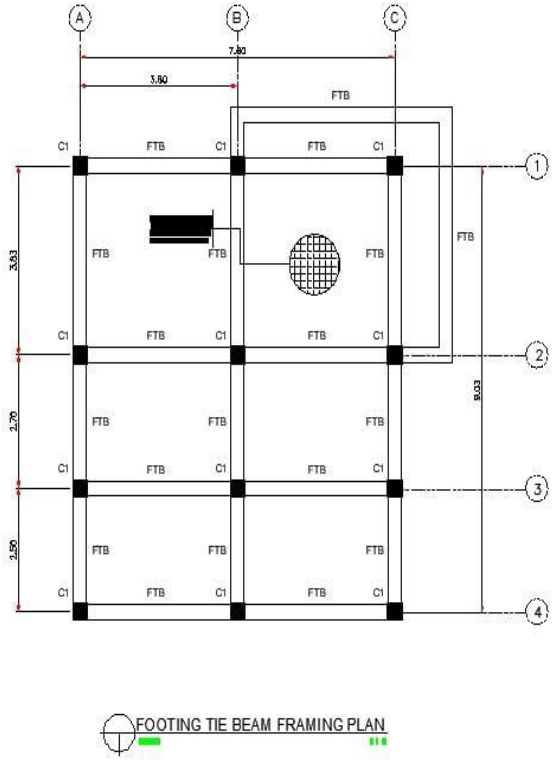
Base Line

For setting out /layout, the most important requirement is to establish a baseline. This is marked on the ground as per site plan requirement with the help of offsets which are taken from the existing road or existing building.

Centre Line

Centre line divides the plan into two equal parts. This can be marked in the field with the help of baseline. This line is very necessary and useful for layout. This line should be transferred to Burjis and be kept up to the completion of foundation work.

'Burjis' and its' Distance Burji or marking pillars are masonry pillars constructed with bricks and cement mortar. These are constructed on both ends of walls /columns and centre line should be marked on the top surface of the burjis with the help of base line. Burji is also constructed for indicating the plinth level of the building. Burjis are very useful for the layout. Accuracy of the foundation can be checked with the help of Burji at any time during construction. Burjis should be kept intact till completion of foundation work.



❖ Types of structure in civil engineering

1. Load bearing structure

A load-bearing structure is the first type of structure in building/civil engineering that has the components of a building that carries and transfers the load to the ground safely.

This type of building structure guarantees the stability of the building and its performance. Commonly walls, columns, beams, and foundation.

There is a transfer of loads from the slab to the walls and from the walls to the foundation.

Slabs >> Walls >> Foundation

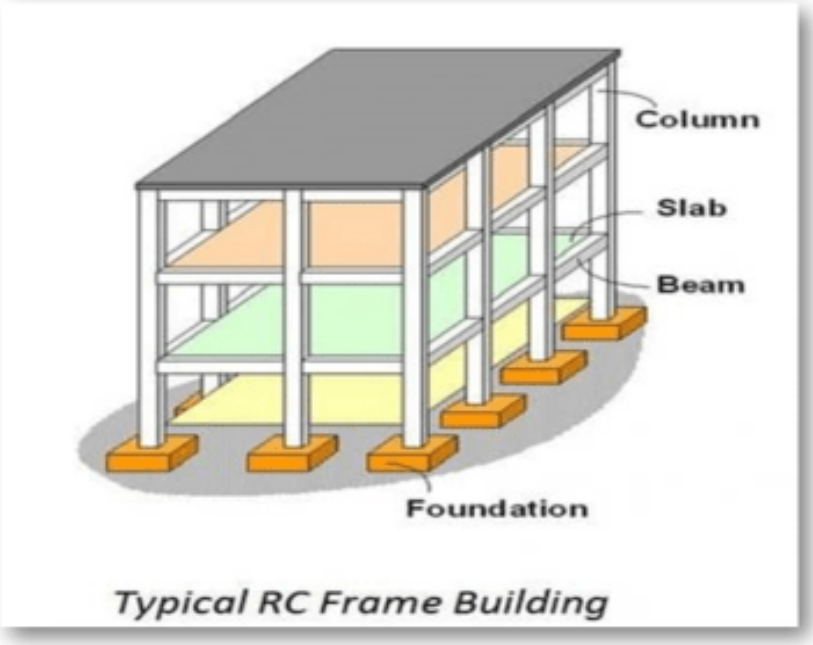
2. Framed structure

Framed structures are another type of building structure in civil engineering. its work is based on the structural elements that transfer the loads from slab to footing.

Slabs >> Beams >> Columns >> Foundation

In the case, of Load bearing,

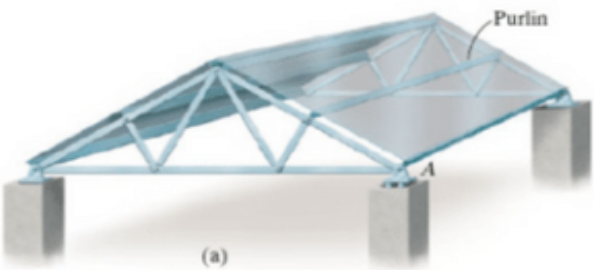
Slabs >> Walls >> Foundation



3. Truss Structure

A truss is an articulated structure composed of straight members arranged and connected in such a way that they transmit primarily axial force. If all the members lie in one plane it is called a plane truss.

A three-dimensional truss is called a space truss. A truss is an efficient way to span long distances which minimizes the amount of material used. This is because the internal loads of the members are induced axially (in the direction of the member) in the form of compression or tension.



❖ Building

Any enclosed or open space with a roof, sidewall, foundation, W.C. & bath, etc., is called a Building**.**

Types of Building

1. Residential Buildings

Residential buildings are those in which at least a sleeping facility is provided for normal residential purposes, with or without cooking and dining facilities (except institutional buildings).
It includes single or multi-family dwellings, apartment houses (flats), lodgings or rooming houses, restaurants, hostels, dormitories, and residential hotels.

2. Educational Building**s**

These include any building used for school, college**,** or day-care purpose**s** involving assembly for instruction, education**,** or recreation and which is not covered by assembly buildings**.**

3. Institutional Buildings

Institutional buildings are used for various purposes such as medical treatment other treatment or care of persons suffering from physical or mental illness, disease or infirmity, care of infants or old age people's care.

The institutional buildings are major including sleeping accommodation for the occupants.
They include hospitals, sanitoria, custodial institutions, or penal institutions like jails, prisons, and mental asylums.

4. Assembly Buildings

These are the building**s** where groups of people meet or gather for amusement**,** recreation, social, religious, political, civil, travel, and similar purposes, such as theatres, motion picture houses, marriage halts**,** town halls, auditoriums, exhibition halls, assembly halls museums**.**

5. Business Buildings

These buildings are used for the transaction of business (other than that covered by mercantile buildings), for the keeping of accounts and records, and for similar purposes, offices, banks, professional establishments, courthouses, and libraries**.**

6. Mercantile Buildings

This type of building is used for selling a small product in which small shops, stores, markets**,** for display and sale of merchandise either wholesale or retail, office, shops,

and **s**torage service facilities incidental to the sale of merchandise and located is known as Mercantile building.

7. Industrial Buildings

Type of building in which raw material process and material fabrication**,** processing or assembly plants**,** laboratories**,** dry cleaning plants**,** power plants**,** pumping stations**,** smokehouses**,** laundries**,** gas plants**,** refineries, dairies**,** and sawmills**.**

8. Storage Buildings

Storage buildings are those in which material **is** stored or sheltering facilities are provided for goods like wares or merchandise**.**

9. Hazardous Building

The hazardous building used for storage of such materials or chemicals that are highly dangerous to humans, or it may pollute the environment**.**

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❖ Carpet Area and Plinth Area

Carpet Area:

It is the covered area of the workable spaces of room at any floor. It is measured between walls to walls within the structure and is the sum of the actual areas of the rooms where you can carpet.

That area which is available for use within a building except the area covered by the walls is called carpet area.

Area included:

The area which is included in carpet area that is given below:

- All rooms (living rooms, bedrooms, dining rooms etc.) in carpet area is included.
- In carpet area the Kitchen and bathrooms is included.
- Stores and balconies are included.

Areas not included:

The area which is not included in carpet area that is given below:

- Common areas (lobbies, gateway etc.) is not included in carpet area.

Plinth Area:

It is the covered built-up area that is measured at the floor level of any floor. It is also known as built-up area and plinth area is the whole area covered by the structure plus internal and external walls. Usually, it is more than 10 to 20% more than carpet area.

Area included:

The area which is included in plinth area that is given below:

- In plinth area whole carpet area is included.
- Internal and external walls is included.

- In plinth area the area of the wall at the floor level, deducting the offsets of plinth. If there are common walls between two structures, half of the area of these walls shall be added in the plinth area.
- The airway for air conditioning and lift well is include in plinth area and landing of lift well.
- Staircase room, Machine room is included.
- Porch is included.
- In plinth area veranda, balconies and parapets is included.

Areas not included:

The area which is not included in plinth area that is given below:

- In plinth area the Common areas is not included.
- Cantilevered porch is not included.
- In plinth area Balcony is involve
- Area of loft is not included.
- Open platform is also not included.
- Spiral staircase and landing are not included.

Difference between Plinth Area and Carpet Area:

There are a lot of difference between culvert and bridge. The difference is given in points below:

S. No	Plinth Area	Carpet Area
1	Also called built up area plinth area is the whole area covered by the structure, including the area covered by the internal and external walls of the building.	That area which is available for use within a building except the area covered by the walls is called carpet area.
2	Plinth area is measured from wall to wall and add the thickness of walls. It means actual area of building plus thickness of wall.	Carpet area is measured from wall to wall within building and translates into the actual floor area which you can carpet
3	Whole area covered by structure	Actual area inside structure.
4	It is 10% to 20% more than the carpet area	It is 10% to 20% less than the plinth area
5	In plinth area balcony is not included	In carpet area balcony is included

Floor area:

The floor area of the building is the total area of the floor in between walls and consists of the floor of all rooms, verandas, corridors etc. The floor area is equal to the plinth area minus area occupied by wall.

❖ Floor Space Index (F.S.I)

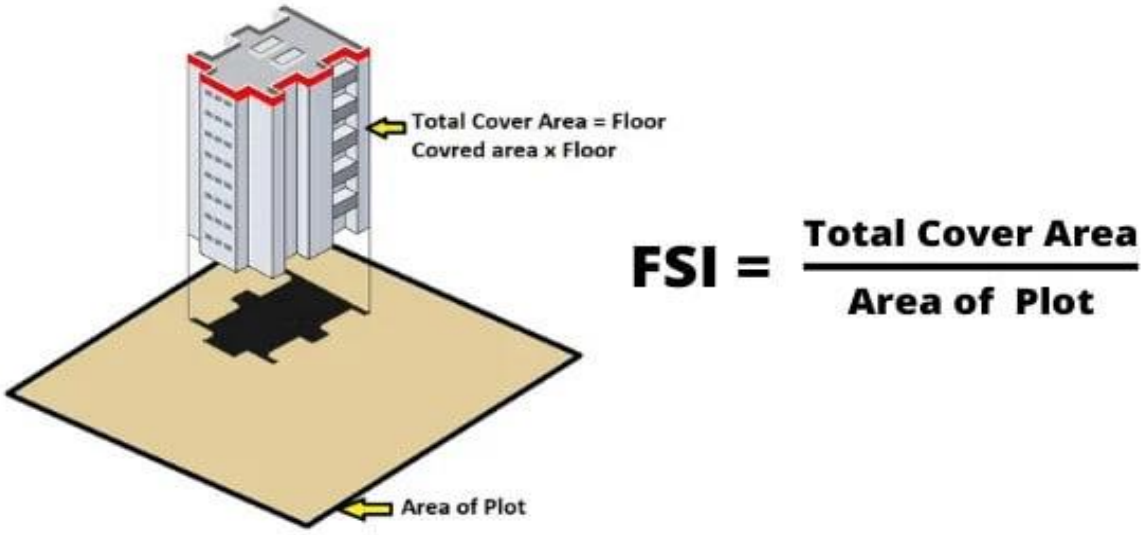
The floor space index means the total area covered by all the floors of the buildings, divided by the total area of the plot on which these buildings are constructed. Also, the floor space index (FSI) is often referred to as the Floor Area Ratio (FAR) in some regions.

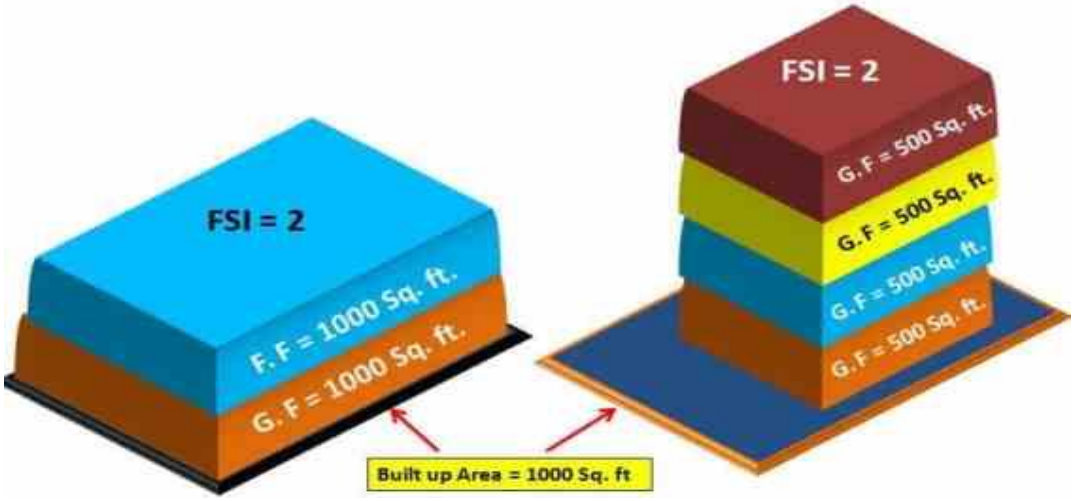
The total floor space area of all the floors by the complete area of the plot is known as the floor space index. It is basically a ratio that helps in deciding how much construction must be done on a piece of land.

Presently, different governing bodies, including Municipal Corporations, Municipal Councils, and other authorities, have different criteria for affixing the maximum FSI for a particular area.

Floor Space Index Formula

Floor Space Index Formula = Total built-up area (total area constructed) divided by Total Area of the Plot





Importance of Floor Space Index

Establishing rules related to floor space index/floor area ratio is of utmost importance for the simple reason to control the population density pressure in an area. Each plot and the area can only consider a certain population density. It increases the burden on the area, especially in terms of basic amenities such as water and wastewater consumption and discharge, respectively.

Further, floor space index/floor area ratio also plays a significant role in the pricing of the building. The buildings or plots with lower floor space index/floor area ratio signify less population density pressure on the area, thus better market value of the property. In comparison to the plot of land with a higher floor space index/floor area ratio.

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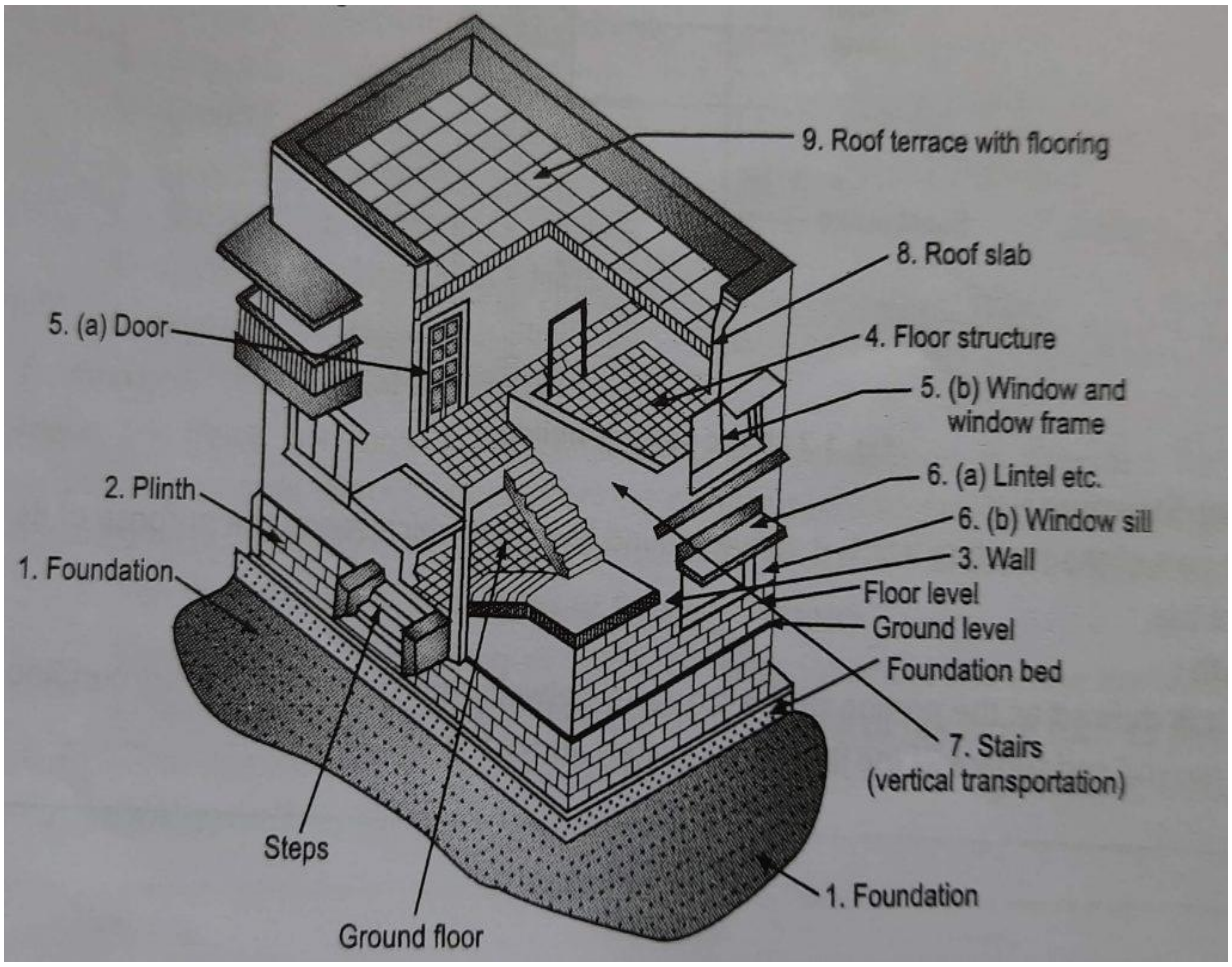
❖ Basic Components of a Building or a Structure and their function

A building is combination of various components. A Civil Engineer should have good knowledge of execution of each component with respect to design layouts given by Architect.



Classification of Component Parts

1. Substructure

It consists of the parts of building below ground level. Function of sub structure is to transmit the load from super structure to the soil.

2. Plinth

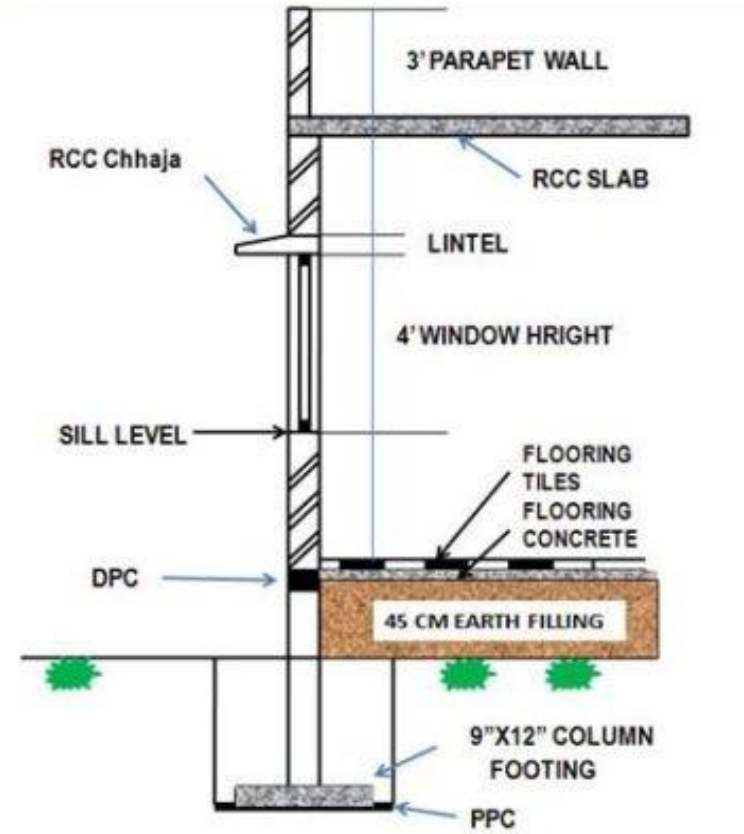
It is connecting level between super and sub structure of building. Located immediately above ground level and below super structure.

3. Super Structure

It is part of structure above ground level whose function is to serve the purpose of building.

Components of Building

The basic Components of a Building Structure are the Footing, Column, Beam, Slab, Lintel, Doors, Windows, Sill, DPC. Earth Filling, Parapet Wall, etc.



1. Foundation

A Foundation is the lowest part of the building structure rested on soil below ground level. All loads of building transferred to foundation through–beam and column arrangement.
Its main function is to distribute the load evenly and safely to the ground.

2. Plinth Level

Plinth level or Plinth is the offset created between ground level and superstructure of the building. It is made by constructing a brick wall from ground level to the ring level of the building.

Its main function is to prevent the entry of moisture from the ground surface to the building superstructure.

3. Earth Filling

An Earth Filling or Soil Filling is done between the plinth walls.

It is essential to fill the open space left between the ground level to the plinth level.

Earth filling must be very well compacted so that the flooring gets a sufficiently hard surface base.

4. DPC (Damp Proof Course)

DPC is a layer of waterproofing material such as asphalt or waterproof cement at the plinth level.
The superstructure walls are constructed above the DPC layer so that no dampness rises from the ground surface into the walls.

In short, DPC prevents the rise of water to superstructures.

If dampness rises in the wall of the superstore, it reduces the strength of the walls and creates unhealthy living conditions.

Also, it created many defects in paint and plaster and ultimately increases maintenance costs.

In the case of Plinth, beams are provided above ground level DPC is not required. Because the plinth Beam itself performs as a DPC layer to restrict the entry of dampness.

5. Flooring

Flooring is an important component of the home. It is one that provided an attractive and pleasant look to the house. Flooring is made by laying tile on it.

6. Walls

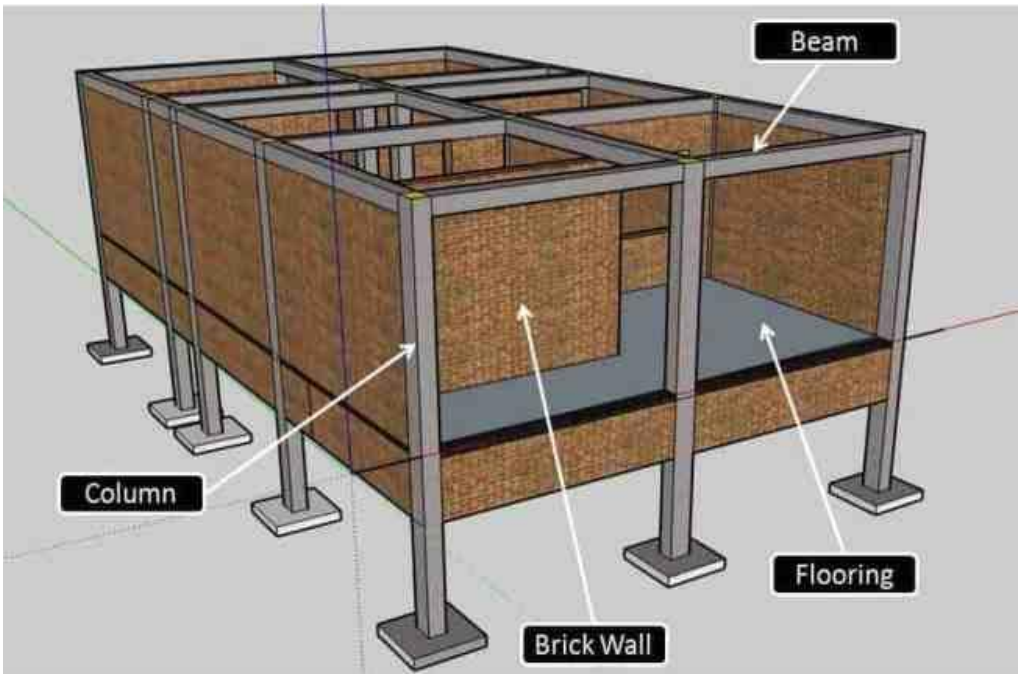
Walls are the vertical component of any structure. It can be constructed using stones, bricks, concrete blocks, etc. The different types of bonds are used for constructing walls.

Bricks walls are essential to enclose the inside area and protect against wind, sunshine, rain, etc. Doors and Windows are provided in the walls for ventilation and access to the building.

7. RCC Column

Columns are vertical members constructed to support any structural frame. Load Coming from the Slab, Beam transferred to column and column transfer load to the footing safely. Building structures may have two types of columns.

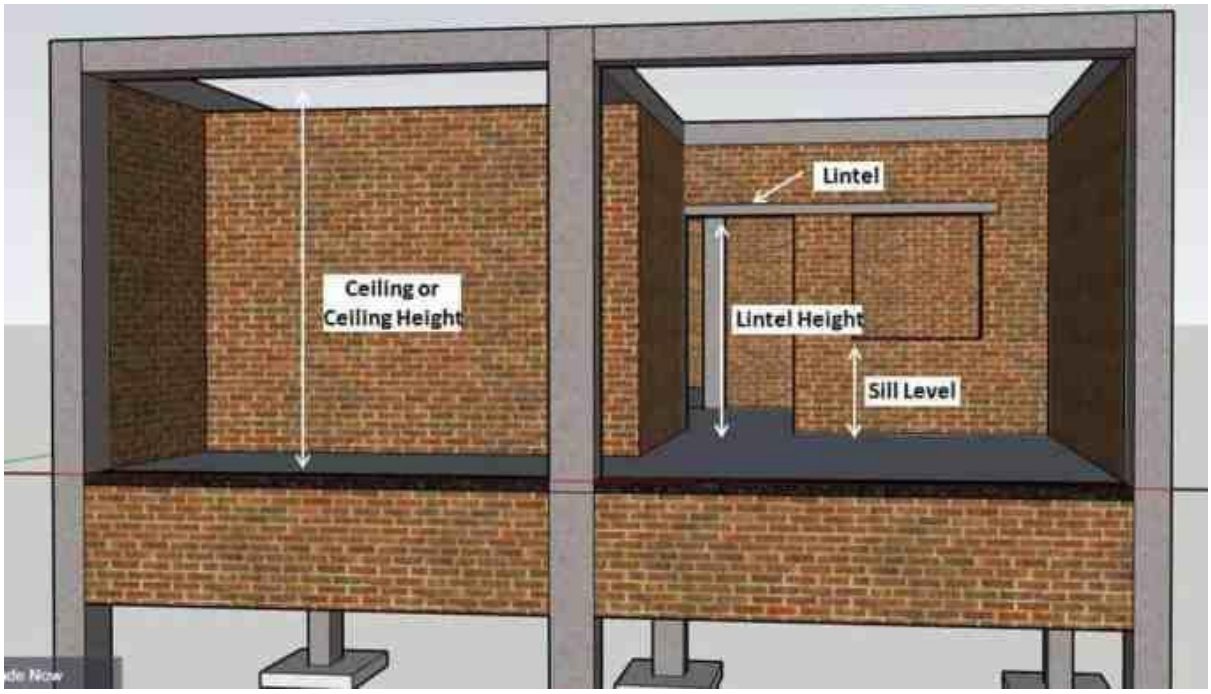
Architectural columns and structural columns. Architectural columns primly used to increase the aesthetic appearance of a building while a structural column takes the load coming from the slab above and transfers it safely to the foundation.



8. Sill Level

It is the base point of any window in the house, or it is a level where windows are placed. The Sill level is very important to ensure evenness in all window levels.

A sill is a height which is ensuring that the proper amount of light enters the house. It also provides easy habitat to look outside through the window without any discomfort. It provides a solid base for window instalment.



9. Lintel

Lintels are constructed from reinforced cement concrete. It is provided above the wall openings like doors, windows, etc. The Lintel takes the load coming on window and door openings.

In Sort, lintel safeguards the door and window from excess load coming from above. In residential buildings, lintel can be constructed from concrete or brick masonry.

10. Ceiling

The ceiling is not a separate part but is the bottom face of any slab. The ceiling is the most important part of any room because it can be decorated to increase the aesthetic appearance. POP (Plaster of Paris) is a material used for making false ceilings.

11. Chhajjas

Chhajjas is also called Weather shed. This is a structure that is constructed above the window projected outside from the window face.

The main function of the weather shed is to restrict the direct entry of rainwater and sunlight. Chhajjas are constructed from reinforced cement concrete.

12. RCC Beam

It is an important component in any frame structure. The beam is a horizontal member which connects columns on both sides.

Its main function is to take the load from the upper structure and transfer it to the column.

Generally, the column to beam connection is called direct support while the beam-to-beam connection is called indirect support.

In most cases, the beam is supported by two columns, and the most rarely used is the cantilever beam.

13. Roof or RCC Slab

The roof is an essential structural component of a building for any structure, which provides protection again environmental factors like sun, wind, and rain.

Generally, all roof rests on sidewalls and required anchoring so that wind and another mechanical impact cannot destroy them.

Roofs may have different shapes, but flat and sloped roofs are more popular. Typically, most roofs are constructed from RCC, stone slabs, tiles, etc.

14. Doors

Doors are the main entry and exit point of any house. Without doors, there will be no security in the house. They are one which separate the one room from others.

Doors have a lock-key facility, so we can lock the house by locking the door and going out freely. They are made of strong materials like steel, wood, and iron. Therefore, they are not easily breakable.

15. Windows

Windows are one of the essential components of any house or building. It is one that allows fresh air and light to enter the house. Without the window, the house becomes a darkroom or jail.
Windows are provided at sill level and their height is extended up to lintel level.

16. Parapet Wall

It is a low–height wall built along the edge of the roof, terrace, walkway, balcony, etc.

Parapet walls can be constructed using different materials like reinforced cement concrete, steel, aluminium, glass, etc. It is generally constructed with a single brick wall.

17. Copping

Coping is a structure that is constructed on top of Boundary walls & parapet walls etc. to protect rainwater directly store on brick masonry walls.

The main function of coping is to drain off rainwater during the rainy season & improve the aesthetics of the structure/wall.

18. Stairs

A stair is a sequence of steps that connects different floors in a building structure. The space occupied by a stair is called as the stairway. There are different types of stairs like a wooden stair, R.C.C stair etc.

19. Plinth Beam

Plinth beam is a beam structure constructed either at or above the ground level to take up the load of the wall coming over it.

20. Plinth

The plinth is constructed above the ground level. It is a cement-mortar layer lying between the substructure and the superstructure.

**********END**********

❖ Building Byelaws

Building byelaws are a set of rules under which construction of a building needs to take place. The rules regulate coverage, height, architectural design, and safety measures to protect buildings against natural disasters such as earthquakes and hazards such as fire as well as structural failures.

The Town and Country Planning Organisation Ministry of Urban Development have devised a circular called "Model Building Bye-Laws - 2016" which states the regulatory mechanism and engineering parameters to keep in mind before starting any construction project in India. Many states in India, experience unauthorised constructions and building hazards due to absence of byelaws. While states can form their own building byelaws, Model Building Bye-Laws - 2016 acts as a standard for construction and a direction for states to devise their own rules for construction.

Building byelaws are necessary to protect any building against noise, fire, earthquakes, structural failures, and any hazardous activity. Many small and medium-sized structures in India don't have building byelaws and are constructed in the absence of any regulatory mechanism with excessive coverage and dangerous development causing a chaotic situation.

Salient features of the Model Building Bye Laws 2016

Several key objectives of building byelaws were focused on the overall safety, security, and transparency of how buildings were constructed in India. Salient features of the 2016 byelaws included the following provisions:

- **Safety & Security:** this included the structural safety, prevention measures, disaster management, etc. related to a building's architectural design.
- **Accessible by all:** these provisions were to make all constructions suitable and safe for children, the elderly, differently abled etc.
- **Eco-friendliness:** this provision was to ensure all buildings incorporated green construction and sustainability mechanisms such as reuse of wastewater, rainwater harvesting, recycling, solar rooftop installations and more.
- **Swachh Bharat Mission:** this provision was to ensure hygienic sanitation facilities for women and the public at large.
- **Adoption of technology:** this provision was to ensure higher adoption of technologies that can increase efficiencies in structural safety, fire safety, disaster management and more.
- **Ease of doing business:** this provision was to ensure that commercial building plans were adequately adhering to compliance requirements and reporting the same regularly.

Aspects Covered by Building Bye Laws

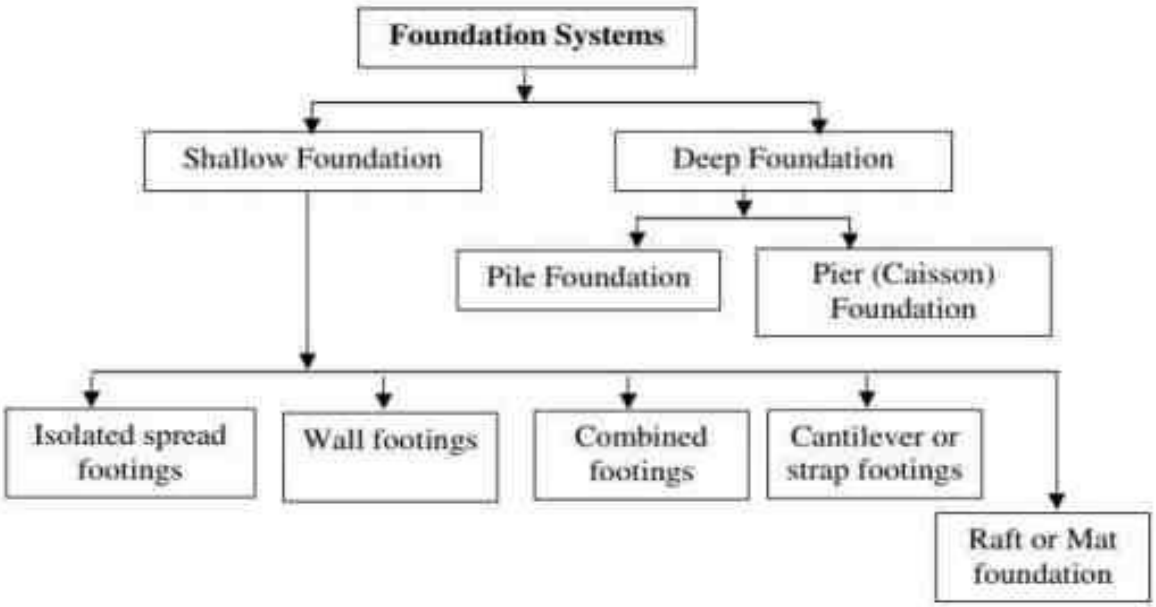
Directions are given in India's building byelaws for the following areas of construction:

- Area covered and application.
- The height of the structure
- Coverage of the structure
- Index of the available floor space
- Density
- Projection and setbacks
- Parking areas.
- Provisions for stairwells and exits in the event of a fire.
- Green places are necessary.
- Areas with lots of light
- Elevators are available.
- sewage treatment plants
- Water is provided.
- The power supply is provided.
- Rainwater collection
- Provisions for safety

❖ Foundation

The Foundation is the main supporting component of any building structure. It is the lowest part of the building which is in contact with soil and transfers all structural load to soil safely. Without a foundation, you may not see skyscrapers building in the world. There are different Types of Foundations used in construction depending upon the nature of the work.

Types of Foundation in Building Construction



Shallow Foundation

When a depth of foundation is less than the width of the foundation then it is known as Shallow Foundation. Generally, a shallow foundation is placed no more than 6 ft depth from the lowest finished floor.

Shallow footing has different types.

Individual footing or Isolated footing

It is the most widely used foundation being used in building construction. This footing may be square, rectangular, or roundabout and is used when the structural load is being carried by columns and used in ordinary buildings of size up to five stories.

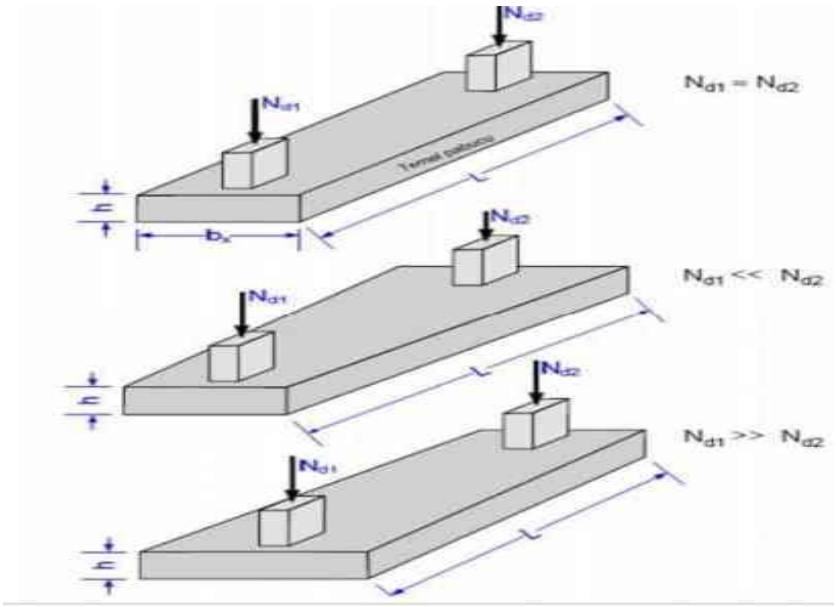
The rectangular shape is chosen when the foundation experiences moments due to the eccentricity of loads or due to horizontal forces.

An isolated footing is used when:

- Structural load is relatively low.
- Columns are placed openly.
- The bearing capacity of the soil is high at a shallow depth.

Combined footing

It is used when two or more columns are placed close enough so that their isolated footings overlap each other. It is of rectangular, square, or trapezoidal shape. A combined footing is made common to more than one column. These are also of different types including slab type, slab, and beam type, rectangular, raft, and strap beam type. The main purpose is to provide uniform distribution of loads for the entire area of the footing.

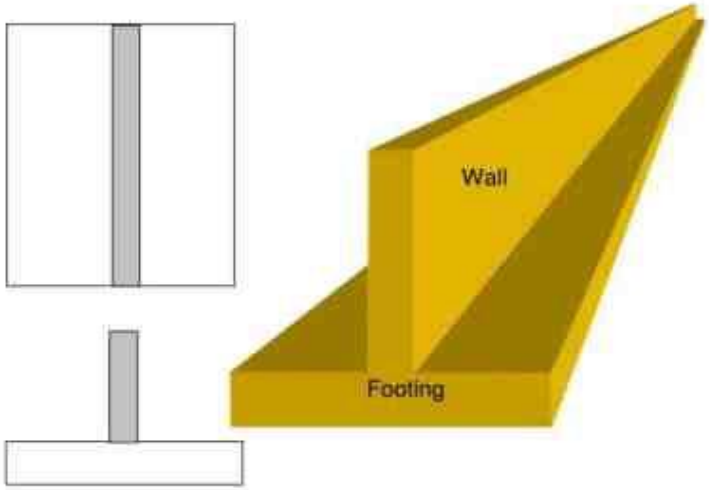


Strip or Wall Footings

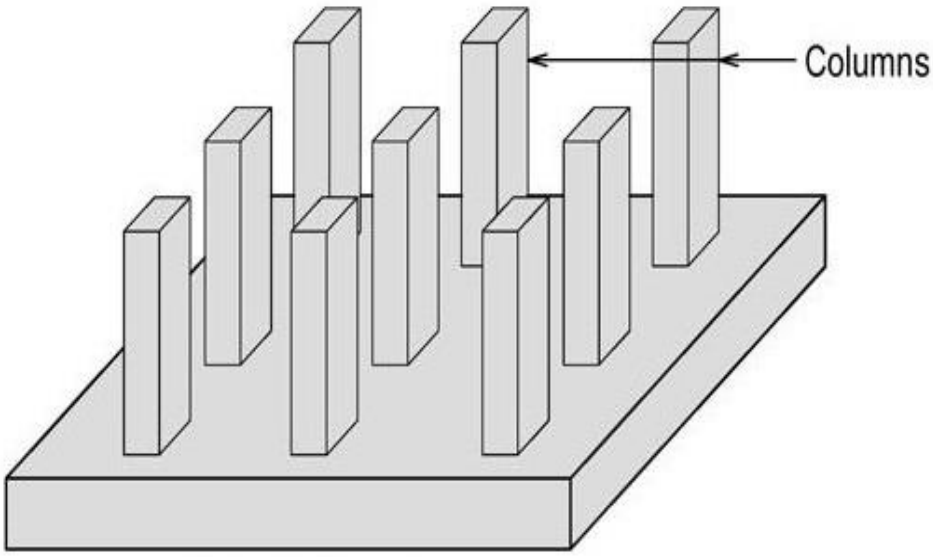
Strip footings act as long strip that bears the weight of a complete wall and are suitable for the buildings where the load is being taken by walls instead of isolated columns.

All footing runs along the direction of the wall. The thickness and size of the footing are specified based on the type of soil at the site. The width of footing is generally kept between 2-3 times the widths of the wall.

In this foundation, the wall footing can be constructed from plain concrete, reinforced concrete stone, brick. The wall footing is best suited for small buildings.



Raft or Mat Foundation



Mat /Raft Foundation

These are used where other types of shallow foundations don't fit suitably. These are used in cases where the structural load is very high. It is being used to prevent differential settlement of individual footings and hence, designed as a single mat of all the load-bearing elements of the structure.

Raft foundation is used:

- For weaker soils and the area over which load is to be spread being large.
- Other kinds of foundations are not feasible.
- Differential settlement is to be prevented.

Cantilever or Strap Footing

Strap footings are like combined footings. Reasons for considering or choosing strap footing are identical to the combined one.

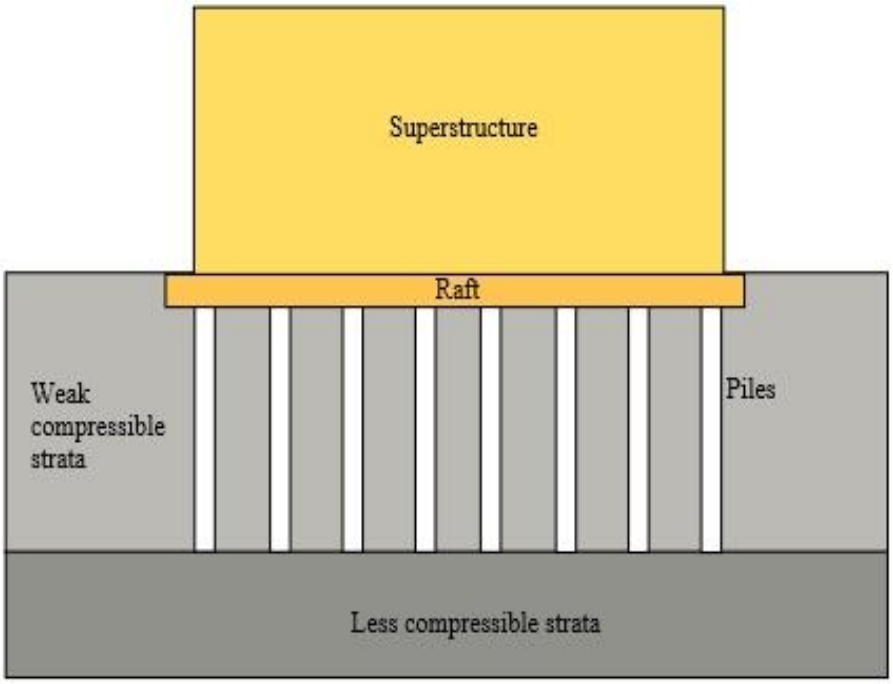
In strap footing, the foundation under the columns is built individually and connected by a strap beam. Generally, when the edge of the footing cannot be extended beyond the property line, the exterior footing is connected by a strap beam with interior footing.

Deep Foundations

A deep foundation is a type of foundation which is placed at a greater depth below the ground surface and transfers structure loads to the earth at depth. The depth to width ratio of such a foundation is usually greater than 4 to 5.

Types of Deep Foundation:
Deep foundation can be classified into the following types:

Pile foundation:
Pile foundation transfers heavy loads from the structure to a hard rock stratum which is located deep down the ground level where shallow foundations cannot be used. It is mainly used to prevent uplift of the structure due to lateral loads such as earthquake and wind forces.



Pile foundation is used when:

- The areas where soil conditions near the ground surface are not suitable for heavy loads.
- The depth of hard rock strata is 5m to 50m deep from the ground surface.

It is further classified on the following basis:

Functions or use of Pile Foundation:

- Sheet piles
- Load bearing piles
- End bearing piles.
- Friction piles
- Soil compactor piles

Materials or construction method:

- Timber piles.

- Concrete piles
- Steel piles
- Composite piles

Pier or Well or Caisson Foundation

Caisson is a watertight structure made of wood, steel, or reinforced concrete that excavates for the foundation of bridges, piers, etc. This type of foundation is mostly used in bridge pier construction and other structures that require foundation underneath water.

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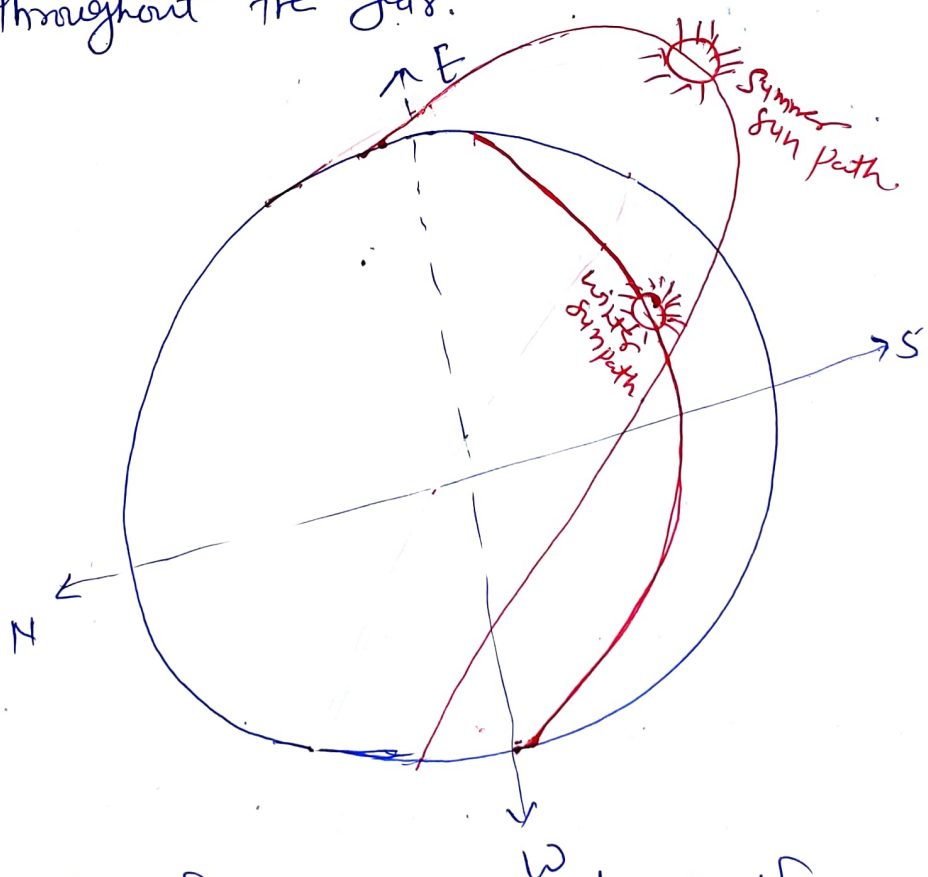
* Concept of Sunlight:-

→ Sunlight is a natural source of light for our buildings during daytime. Sun shining into a building provides free heat and natural light.

→ For the maximum utilization of sun light, we have to study about the sun path diagram.

→ Sunpath diagram can tell you a lot about how the sun will impact our site of building throughout the year.

→ Sunpath diagram provides a broader overview of sun on a site as they map the path of the sun across the sky at different time during the day throughout the year.



→ Sun path refers to the apparent significance of seasonal and hourly positional changes of the sun as the earth rotates and orbits around the sun.

→ The position of sun is in Northern hemisphere and near to earth surface during summer season. But in winter season position of sun shifted to Southern hemisphere and away from the surface of earth.

→ By studying the sunpath diagram of a place one can identify the optimal orientation of buildings so that we can utilize maxm sunlight during winter and minm during summer.

* Concept of ventilation:-

→ ventilation moves outdoor air into a building or a room, and distribute the air within the building or room.

→ ventilation refers to any system that allows air to flow and move between the inside and outside of a property, providing fresh air for the occupants.

⇒ Purpose of ventilation:-

1). Provide a Continuous supply of fresh ~~air~~ outside air.

2). Removal of dilute airborne Contaminants.

3) Reduce potential fire or explosion hazards.

4). Maintain temp and humidity at Comfortable levels.

5). Controlling moisture.

6). Controlling temperature

7) improving health of the residents.

⇒ Types of ventilation:-

1) Natural ventilation:-

- Utilizing resources that are already present in nature to provide ventilation is known as Natural ventilation in building.
- By installing windows across various areas of the walls to allow for wind-driven ventilation.
- Windows can be constructed with the windward side in mind to help with natural ventilation.

2) mechanical ventilation:-

- When natural ventilation is not possible then mechanical ventilation is used.
- Mechanical ventilation methods include demand control and local exhaust ventilation.

⇒ Effect of poor ventilation:-

→ Building have poor ventilation may have condensation issues.

→ Inadequate ventilation might cause health risks for occupant of the buildings.

→ poor ventilation in building can lead to the growth of mold plant on wall.

→ An increase in humidity & mold plant growth in the wall can result in damage to the walls.

* National Building Code of India (NBC)

→ The national building code (NBC) is a document that provides guidlines for construction of structures like residential, mercantile, Institutional, educational, Commercial, assembly, storage spaces or even hazardous buildings.

→ It is Important to follow guidlines that are ment to protect the overall health of the Construction and ensure the health and safety of the public and the residents.

* R.C.C (Reinforced cement Concrete) :-

→ Reinforced cement Concrete (R.C.C) is the Combination of ordinary Concrete with the reinforcement to increase its Compressive and tensile strength to a great extent.

→ ordinary Concrete is a mixture of cement, Sand, Aggregate, water and admixture.

Purpose of Reinforcement cement Concrete :-

Since Concrete has a very high Compressive strength, but it is low in tensile strength.

Thus, when only the Compressive force are acting on the Concrete surface, then their is no need of using reinforment in it.

But where tensile force are also involved, as in, beams and slabs, there is a very high risk of its failure when ordinary concrete is used. Hence Reinforcement is added in case of tension.

Advantages of reinforced Concrete (RCC):-

1) Structures made from Reinforced concrete are durable.
2). It has high compressive strength (due to concrete).
3). It has a high tensile strength (due to reinforcement).
4) It is resistant to fire and other climate changes.
5) Easily available.
6) It can be molded in any shape.

Uses of RCC:-

1). RCC forms the major structure of a building structure. All main load-bearing components of the building are made with RCC.
2). RCC is also used for construction of storage structures like water tank, Dams etc.
3). It is used for the construction of large infrastructure project like bridges, Retaining walls etc.
4). RCC is also used for precast element casting like railways sleepers, Electric poles etc.
5). It is used for constructing tall structures like multistory buildings, towers etc.
6). RCC is used for paving work for Roads and canal.

* Cement:-

→ Cement is a binder, a chemical substance used for construction that sets, hardens and adheres to other materials to bind them together.

→ Cement is finely ground particles that, when mixed with water, set to a hard mass.

⇒ Uses of cement:-

1) Cement is used in mortar for plastering, masonary work, pointing etc.
2) Cement is used for making joints for drains and pipes.
3) Cement is used for water tightness of the structure.
4) Cement is used in concrete for laying floors, roof, and constructing lintels, beams and stairs etc.
5) It is used for precast pipes manufacturing, piles, fencing post etc.
6) It is used in civil Engineering construction work.
7) Cement is used in the preparation of waterright floors.

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